Ashraf Sada

Full Stack Developer & Systems Engineer

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Summary

Full Stack and Systems Engineer with 12+ years of experience across web, desktop, and cross-platform mobile development. Comfortable owning the full stack from low-level systems work in Zig to production ASP.NET Core services and modern JavaScript front-ends (React, Solid, Blazor). Ships across Windows and Linux with strong database design (SQL Server, PostgreSQL) and CI/CD automation.

Highlights

- Architected a multi-tenant ERP (72K SKUs) with ML-driven forecasting, cutting PO cycle time by 80%.
- Led a 4-dev team building a supply chain ERP replacing legacy VB6, cutting time-to-market from 3 days to 24 hours.
- Designed a HA medical scheduling platform in Zig with PostgreSQL replication and HAProxy load balancing.

Technical Skills

- **Languages:** C#, Zig, Elixir, JavaScript, Python, Bash, PowerShell
- **Back-end / APIs:** ASP.NET Core, ASP.NET WebForms, REST, CQRS, DDD, Phoenix, LiveView
- **Front-end:** React, Next JS, Solid JS, Blazor, Vanilla JS, CSS, SCSS, Tailwind CSS
- **Mobile / Cross-Platform:** .NET MAUI, Flutter
- **Systems / OS:** Windows, Linux, Docker, systemd, HAProxy, FFI, Concurrency, Byte Management (memory hierarchy)
- **Networking:** HTTP/HTTPS, WebSockets, TCP/UDP sockets, IP/Subnets, VPCs, network architecture
- **Databases:** T-SQL (SQL Server), PostgreSQL, SQLite
- **DevOps / Automation:** GitHub Actions, Forgejo Actions, Azure DevOps, Docker, Nginx, YAML
- **Testing:** xUnit, Playwright, pytest

Work Experience

Freelance Software & Systems Engineer

Aug 2024 - Present | Freelance, Amman, Jordan

Parcel Delivery System (PDS) | C#, ASP.NET Core, Blazor, .NET MAUI

- Designed and delivered a full-stack parcel delivery platform replacing manual Excel-based operations: ASP.NET Core Web API back-end, Blazor web front-end, and .NET MAUI mobile app for field agents (shipment tracking, routing, client management).
- Serves 1,500+ clients, consolidating dispatch, tracking, and billing into a single system.
- Deployment: self-hosted on Linux, Nginx reverse proxy, CI/CD via GitHub Actions.

ERP System | C#, ASP.NET Core, Blazor, ML.NET

- Architected a multi-tenant ERP system (72K SKUs) using Domain-Driven Design, Clean Architecture, and CQRS with a Blazor front-end.
- Implemented ML.NET sales forecasting (linear regression) that automatically updates target purchase quantities based on price changes and demand trends, driving automated purchase-order generation.
- Reduced PO and forecasting cycle time by 80%, improving purchase quantity accuracy and costing precision.
- Deployment: Azure cloud SaaS, CI/CD via GitHub Actions.

Full Stack Developer
Jan 2022 - Jul 2024 | Mafateeh Group IT and Media Solutions, Amman, Jordan

Company Website Migration | PHP to ASP.NET Core, Blazor WASM

- Migrated company website (Home, About, Services, Contact) from PHP to ASP.NET Core with Blazor WASM front-end.
- Improved page load performance, modernized UI/UX, and added client-side validation with anti-probing protections.
- Deployment: self-hosted on Windows Server, CI/CD via GitHub Actions (YAML pipelines).

Digital Marketing CMS | ASP.NET Core, Blazor

- Built a maintenance management system for Digital Out-of-Home (DOOH) advertising: monitors street digital billboards by reading error sensors, tracks and dispatches maintenance jobs.

Computerized Maintenance Management System (CMMS) | ASP.NET Core, Blazor

- Designed a custom CMMS for a leading offline marketing company in Jordan: links each equipment model to its schematic diagram, reads fault codes, and suggests most probable cause with recommended actions.
- Manages spare parts inventory directly tied to equipment models, reducing downtime by eliminating manual lookup.
- Built a fully responsive technician portal (PWA) used by ~20 field technicians, available 24/7.

Technical Support Management System (TSMS) | React, Next.js, ASP.NET Core

- Proposed and built an internal ticketing system for managing client technical support: ticket tracking, priority-based categorization, and resolution workflow.
- Handles 30+ client accounts.

Online Instructor (Independent)

Apr 2020 - Dec 2021 | Remote

- Taught university students online: Java, C#, web layout, CSS, and full-stack fundamentals.
- One student (top 10 graduate, Software Engineering) was trained on practical web development and secured a job post-pandemic.

Team Lead, Full Stack Developer

Nov 2018 - Mar 2020 | TATWIR Foods, Amman, Jordan

Supply Chain & Sales Distribution ERP | C#, ASP.NET Core MVC, SQL Server

- Led a team of 4 junior/mid-level developers to design and build a full ERP system replacing a legacy VB6 application: automated supply chain operations and sales distribution across 3 warehouses and 20 distribution points (40+ SKUs).
- Delivered semi-automated order-to-delivery workflow, deep insight reporting and analysis, tax-compliant invoicing, and accurate cash management.
- Replaced manual data entry with hand-held device order capture and on-site sales invoice printing; sales data syncs directly to the system, significantly reducing input errors.
- Eliminated DB corruption and data leakage by enforcing null constraints and replacing the legacy single-machine architecture with a fully web-based system accessible from hand-helds and by managers remotely.
- Set up CI/CD pipeline (Azure DevOps + Docker), cutting time-to-market from 3 days to 24 hours.
- Coordinated system requirements between operations stakeholders and the dev team; ran 2-week Scrum sprints.
- Mentored junior developers on front-end and back-end development practices.

Software Engineer / Product Manager

Oct 2016 - Nov 2018 | ProActive Business Solutions, Abu Dhabi, UAE

Zakat Fund UAE (Government Contract) | C#, ASP.NET, Xamarin

- Participated in a 7-developer team delivering the web application and Xamarin mobile app under a 12-month government contract; built various sections in compatibility with the work-flow engine (developed by a 3rd-party sub-contractor).

ERP Feature Development | C#, ASP.NET Core

- Developed new features for ProActive's ERP platform used by multiple private-sector clients across UAE, especially construction companies; leveraged prior accounting background to translate domain-specific requirements into technical solutions.
- Independently designed and developed a real estate management system targeting the UAE market.
- Added multi-dimensional analysis to the financial reporting module.
- Designed and developed a Letters of Credit (LC) management module.
- Designed a Fixed Assets management module with improved depreciation tracking.

Product Management | C#, ASP.NET Core

- Acted as product manager: met with stakeholders and end users, translated business requirements into technical specs for the dev team lead and operations manager.

Software Engineer
Sep 2014 - Oct 2016 | Yamar Business Solutions Development, Amman, Jordan

Car Dealership Estimation Tool | C#, WinForms

- Designed and built a desktop app for a car dealership that scrapes live market prices in Jordan and calculates optimal buying prices using a proprietary formula weighted against vehicle specs and options; also handles used-car and trade-in valuation.

Ministry of Labor Website (Government) | C#, ASP.NET WebForms

- Worked on a team of 3 to extend the existing Ministry of Labor web application with new features, working within the established codebase.

Personal Projects

Xevora - Medical Appointments Platform | *Elixir, Phoenix LiveView, PostgreSQL, Flutter (In Progress)*

- **High-Scale Concurrency Architecture:** Engineered a fault-tolerant backend on the BEAM VM, optimized to handle **~45,000 daily transactions** across **1,300+ medical centers**. Leveraged **OTP Supervision Trees** and **lightweight processes** to ensure high availability and graceful degradation under peak load.
- **Real-Time User Experience with Phoenix LiveView:** Replaced traditional SPAs with **Phoenix LiveView** to deliver reactive, server-rendered UIs. Utilized **`Phoenix.PubSub`** for instant, WebSocket-driven appointment updates and real-time conflict resolution, reducing client-side overhead and improving time-to-interactive.
- **Asynchronous Notification Pipeline:** Designed a non-blocking notification system using **Elixir `Task` supervisors** and background workers to manage real-time alerts for patients and staff. Ensured reliable delivery of critical updates without blocking the main request thread or degrading API latency.
- **Robust Data Integrity & Scalability:** Implemented a high-availability **PostgreSQL** cluster (Primary + 2 Read Replicas via repmgr) with **Ecto connection pooling**. Enforced strict business logic and data validity using **Ecto changesets** and **guard clauses** at the application boundary, ensuring zero-corruption in medical scheduling data.
- **Decoupled Multi-Client Architecture:** Structured the backend using **Phoenix Contexts** to decouple business logic from transport layers, providing a consistent, type-safe API for both the **LiveView web interface** and **Flutter mobile app**, facilitating parallel development and easier maintenance.

Education

- **Pioneers Academy** | Certificate in Software & Web Developer (Pearson Foundation certified) | Aug - Sep 2014
- **Applied Science University** | B.A. In Accounting | Jul 1998

Languages

- Arabic (native)
- English (fluent)
- Spanish
- Turkish